

[HW 0-3]

- Complex Manipulation

↳ Cartesian form ($a + bi$)

↳ Polar form ($re^{i\theta}$)

↳ Algebra in $\mathbb{C}$

↳ Complex Conjugates

- Geometric Series (finite)

$$\sum_{k=0}^n e^{i k \pi / 8} = \frac{1 - (e^{i \pi / 8})^{n+1}}{1 - (e^{i \pi / 8})}$$

- Complex Analysis

↳ Periodicity

↳ Linear dependence

↳ Rotation ($e^{i\theta + \phi}$)

- Taylor Series

$$1 + i \frac{\pi}{2} + \frac{(i \frac{\pi}{2})^2}{2!} \dots$$

$$= e^{ix} = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

- Aliasing

↳ Given some s_s ,

$$| \text{freq} - n \cdot s_s | \bmod (s_s/2)$$

- Trig Identities

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$\sin(x+y) = \sin x \cos y + \cos x \sin y$$

$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$

- Quantization

↳ SNR w/ uniform error distribution

$$-dB = 20 \cdot \log_{10} \left(\frac{s_2}{s_1} \right)$$

↳ Root mean square of specified samples

- Phasors

$$\text{↳ } f(t) = e^{i \omega t}, f(t): \text{Hz}$$

↳ Frequency: $\omega/2\pi$

↳ Phasor Sum

↳ Sampling Freq. (Sample rate) $s_s = 1/T_s$

- Discrete Fourier Basis

$$\text{↳ } C_k = \frac{1}{N} \sum_{n=0}^{N-1} x_n e^{-i 2\pi k n / N}, C_k: \text{Fourier Coeff.}$$

$$\text{↳ } \sum_{k=0}^{N-1} x_k e^{i 2\pi k n / N}$$

[Feed-forward Filters]: delayed phasor

$$\hookrightarrow Y_t = X_t + a X_{t-1} \rightarrow Y_t = e^{j\omega} (1 + a e^{-j\omega})$$

$$\hookrightarrow \text{Frequency Response: } H(\omega) = 1 + a e^{-j\omega}$$

$$\hookrightarrow \text{Phase Response: } e^{j\omega}$$

$$\hookrightarrow \text{Phase: } \omega$$

$$\hookrightarrow \text{Magnitude Response: } |H(\omega)| = \left[(1 + a \cos(\omega))^2 + (a \sin(\omega))^2 \right]^{1/2}$$

$$= [1 + 2a \cos(\omega) + a^2]^{1/2}$$

$\hookrightarrow$ Impulse Response

$$\hookrightarrow S_t = \begin{cases} 1, & t=0 \\ 0, & \text{else} \end{cases}$$

$\hookrightarrow$ Transfer Function, delay operator z^{-1}

$$H(z) = 1 + a z^{-1}$$

$$Y = X + a z^{-1} X$$

$$Y/X = 1 + a z^{-1}$$

$\hookrightarrow$ Palindromic phase Response

$\hookrightarrow$ Inverse Comb Filter

$$Y_t = X_t - R^L X_{t-L} \quad \text{zeros @ } z^L = R^L$$

$$H(z) = 1 - R^L z^{-L} \quad \text{Poles @ } z=0, e^{j\omega} = 0$$

$$= \frac{z^L - R^L}{z^L}$$

[Reson Filters]

2 Pole Reson

$\hookrightarrow$ Max Values

$$\frac{1}{(1 - R e^{j\omega} z^{-1})(1 - R e^{-j\omega} z^{-1})}$$

$$H(z) = \frac{1}{(z - R e^{j\omega})(z - R e^{-j\omega})}$$

[Feedback Filters]

$$Y_t = X_t + Y_{t-1}$$

1 [Reson]

↳ Pole Radius R

↳ Gain Factor

↳ Pole angle

~~1 [Phase Shift]~~

1 [Comb]

↳ $H(z) =$

[Law of Cosines]

$$Y_t = A_0 X_t + 2R \cos \theta Y_{t-1} - R^2 Y_{t-2}$$

↳ Bandwidth & Resonant Freq.

$$B: [R \approx 1 - B/2]$$

↳ Poles on unit circle

$$z = e^{j\theta}, z^* = e^{-j\theta}$$

1 [Plucked-String]

↳ as whole #

$$L \frac{s}{s} = L + 1/2 + \delta$$

$$Y_t + a Y_{t-1} = a X_t + X_{t-1}$$

$$H(z) = \frac{Y_t}{X_t} = \frac{a + z^{-1}}{1 + a z^{-1}}$$

$$Y(1 + a z^{-1}) = X(a + z^{-1})$$

$$H(z) = \frac{a + z^{-1}}{1 + a z^{-1}}$$

↳ Phase Response, allpass component

$$\theta_\omega = 2\theta$$

$$\tan^{-1} x \approx x$$

$$2 \tan^{-1} \left(\frac{a-1}{a+1} \tan \left(\frac{\omega}{2} \right) \right) \quad \theta_\omega = 2 \left(\frac{a-1}{a+1} \omega \right)$$

[Periodic Variations] $\omega_0 = 2\pi/T$ [phasor] $f(t) = e^{isk\omega_0 t}$

↳ Inner Product

$$\frac{1}{T} \int_0^T f(t) \overline{g(t)} dt$$

↳ Forward Fourier

$$\frac{1}{T} \int_0^T f(t) e^{-isk\omega_0 t} dt = C_k$$

↳ Inverse Fourier

$$f(t) = \sum_{k=-\infty}^{\infty} C_k e^{isk\omega_0 t} = C_0 + 2 \sum_{k=1}^{\infty} [a_k \cos(k\omega_0 t) - b_k \sin(k\omega_0 t)]$$

$$C_k = a_k + b_k i$$

[Square Wave

$$f(t) = \begin{cases} 1 & 0 \leq t < T/2 \\ -1 & T/2 \leq t < T \end{cases}$$

$$C_k = \frac{s}{k\pi} [(-1)^k - 1]$$

$$a_k = 0$$

$$b_k = \frac{1}{k\pi} ((-1)^k - 1)$$

$$b_k = \begin{cases} 0 & \text{even } k \\ -\frac{2}{k\pi} & \text{odd } k \end{cases}$$

$$\int e^{at} = \frac{1}{a} e^{at} + C$$

$$\int e^{sat} = \frac{1}{sa} e^{sat}$$

$$1/s = -s \quad 1/-s = s$$

$$f(t + kT) = f(t)$$

$$f(t)_{\text{square}} = 2 \cdot \sum_{\substack{k=1 \\ \text{odd } k}}^{\infty} \frac{2}{k\pi} \sin(k\omega_0 t)$$

$$\frac{4}{\pi} \sum_{\substack{k=1 \\ \text{odd } k}}^{\infty} \frac{1}{k} \sin(k\omega_0 t)$$